# CS & IT ENGINEERING

2025

## Database Management System

### Query Languages

DPP-03 Discussion Notes




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[1 Mark]

#Q. Consider the following relational schema

actor (insta_id, name, language, age) Note: unique name of each actor.

movie (movie_id, title, year, director_id) Note: title is unique for each movie.

acts_in (insta_id, movie_id, character_name )

director (director_id, name, language) Note: unique name of each director.

Write a TRC query to retrieve details of all movies that were released in 2010. The output schema should be the same as that of the movie table.

$\{t \mid t \in movie \wedge t.year = 2010\}$

[1 Mark]

#Q. Consider the following relational schema

actor (insta_id, name, language, age) Note: unique name of each actor.

movie (movie_id, title, year, director_id) Note: title is unique for each movie.

acts_in (insta_id, movie_id, character_name )

director (director_id, name, language) Note: unique name of each director.

Write a TRC query to retrieve details of all actors that are not in their thirties (i.e., age<30 or age>39). The output schema should be the same as that of the actor table.

$$\{ t \mid t \in actor \wedge (t.age < 30 \vee t.age > 39) \}$$

[1 Mark]

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actor (insta_id, name, language, age) Note: unique name of each actor.

movie (movie_id, title, year, director_id) Note: title is unique for each movie.

acts_in (insta_id, movie_id, character_name )

director (director_id, name, language) Note: unique name of each director.

Write a TRC query to retrieve the names of all directors.

$\{t.name \mid t \in director\}$

[1 Mark]

#Q. Consider the following relational schema

actor (insta_id, name, language, age) Note: unique name of each actor.

movie (movie_id, title, year, director_id) Note: title is unique for each movie.

acts_in (insta_id, movie_id, character_name )

director (director_id, name, language) Note: unique name of each director.

Write a TRC query to retrieve the names of all "Telugu" language directors.

$\{ t.name \mid t \in director \wedge t.language = \text{'Telugu'} \}$

[1 Mark]

#Q. Consider the following relational schema

actor (insta_id, name, language, age) Note: unique name of each actor.

movie (movie_id, title, year, director_id) Note: title is unique for each movie.

acts_in (insta_id, movie_id, character_name )

director (director_id, name, language) Note: unique name of each director.

Write a TRC query to retrieve the name of each actor together with the titles of the movie he/she has performed in.

$\{ t \mid \exists a \in actor, \exists m \in movie, \exists n \in acts_in ( a.insta_id = n.insta_id \wedge n.movie_id = m.movie_id \wedge t.name = a.name \wedge t.title = m.title) \}$

[2 Marks]

#Q. Consider the following relational schema

actor (insta_id, name, language, age) Note: unique name of each actor.

movie (movie_id, title, year, director_id) Note: title is unique for each movie.

acts_in (insta_id, movie_id, character_name )

director (director_id, name, language) Note: unique name of each director.

Write a TRC query to retrieve the names of all actors that have played the character of "Ravan".

$\{ t.name \mid t \in actor \wedge \exists n \in acts_in\ (t.insta_id = n.insta_id \wedge n.character_name = \text{'Ravan'}) \}$

[2 Marks]

#Q. Consider the following relational schema

actor (insta_id, name, language, age) Note: unique name of each actor.

movie (movie_id, title, year, director_id) Note: title is unique for each movie.

acts_in (insta_id, movie_id, character_name )

director (director_id, name, language) Note: unique name of each director.

Write a TRC query to retrieve the names of all actors that have played the character of "Ravan", together with the year the corresponding movies were released.

$\{t \mid \exists a \in actor \wedge \exists m \in movie \wedge \exists n \in acts_in (a.insta_id = n.insta_id \wedge n.movie_id = m.movie_id \wedge n.character_name = 'Ravan' \wedge t.name = a.name \wedge t.year = m.year)\}$

[2 Marks]

#Q. Consider the following relational schema

actor (insta_id, name, language, age) Note: unique name of each actor.

movie (movie_id, title, year, director_id) Note: title is unique for each movie.

acts_in (insta_id, movie_id, character_name )

director (director_id, name, language) Note: unique name of each director.

Write a TRC query to retrieve all actors that acted in movie with title "Bahubali". The output schema should be the same as that of the actor table.

$\{t \mid t \in actor \wedge \exists m \in movie\ [\exists n \in acts_in\ (t.insta_id = n.insta_id \wedge n.movie_id = m.movie_id \wedge m.title = 'Bahubali')]\}$

[2 Marks]

#Q. Consider the following relational schema

actor (insta_id, name, language, age) Note: unique name of each actor.

movie (movie_id, title, year, director_id) Note: title is unique for each movie.

acts_in (insta_id, movie_id, character_name )

director (director_id, name, language) Note: unique name of each director.

Write a TRC query to retrieve the names of all actors that have performed in a movie directed by "Anurag Kashyap".

$\{ t.name \mid t \in actor \wedge \exists d \in director \wedge \exists m \in movie \wedge \exists x \in acts_in$
$(t.insta_id = x.insta_id \wedge x.movie_id = m.movie_id \wedge$
$m.director_id = d.director_id \wedge d.name = 'Anurag\ Kashyap') \}$

$t = r_1$
$t : r_2$
$r_1 = r_2$

[2 Marks]

#Q. Consider the following relational schema

actor (insta_id, name, language, age) Note: unique name of each actor.

movie (movie_id, title, year, director_id) Note: title is unique for each movie.

acts_in (insta_id, movie_id, character_name )

director (director_id, name, language) Note: unique name of each director.

Write a TRC query to retrieve the titles of all movies in which Amitabh and Jaya have co-acted.

$\{ t.title \mid t \in movie \wedge \exists a_1 \in actor \wedge \exists a_2 \in actor \wedge \exists r_1 \in acts_in$
$\wedge \exists r_2 \in acts_in ( a_1.insta_id = r_1.insta_id \wedge a_1.name = 'Amitabh'$
$\wedge a_2.insta_id = r_2.insta_id \wedge a_2.name = 'Jaya \wedge$
$t.movie_id = r_1.movie_id \wedge t.movie_id = r_2.movie_id ) \}$

# THANK - YOU